Quantum Simulation of the Electronic Structure of H_2 Using Variational Quantum Algorithms

Presented by

Raphael MARQUES ARAUJO

Youssef BENADDI

Tony PANSERA

Iris VERMEIL

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1 Introduction

If quantum computation has broad potential applications, one of the most promising is quantum chemistry. Fundamental theoretical questions about the behavior of electrons in the electronic cloud arose after the planetary model of the atom was shown to be incorrect.

We discovered that, like all quantum particles, electrons cannot be described deterministically. Instead, their behavior must be described probabilistically. Although we cannot determine their exact position and momentum simultaneously, we can predict their state and the probability of finding them in a given region of space, called spin-orbitals, using the quantum state of the system.

The main difficulty is that, even though the Schrödinger equation is well known and methods exist to solve it, solving it exactly for molecules is extremely challenging. The Hamiltonian, which describes the total energy and interactions in the system, contains a very large number of terms due to electron–electron and electron–nucleus interactions. As the number of particles increases, the computational cost grows exponentially due to the massive number of possible electronic configurations, making exact solutions practically impossible for classical computers.

Consequently, one promising alternative is to reformulate the problem in a way that allows it to be simulated using quantum computation, which is naturally suited for such combinatorial problems through qubit superposition. The primary goal of this project is to use this technology to approximate the ground state energy, or the smallest amount of energy in a system by identifying the most optimized electronic configuration. By implementing a framework using PySCF and Qiskit, we aim to simulate the H_2 molecule and evaluate how different mappings and optimizers impact the precision of these results.

2 Shrodinger's equation and hamiltonian

In a nutshell, the electronic Hamiltonian describes all interactions involving the electrons in a molecular system. Consequently, we can decompose it into several physically meaningful contributions:

$$\hat{H} = \hat{T}_e + \hat{T}_N + \hat{V}_{ee} + \hat{V}_{NN} + \hat{V}_{eN} \tag{1}$$

where:

- $\hat{T}_e$ is the electronic kinetic energy operator,
- $\hat{T}_N$ is the nuclear kinetic energy operator,
- $\hat{V}_{ee}$ represents the electron–electron Coulomb repulsion,
- $\hat{V}_{NN}$ represents the nucleus–nucleus Coulomb repulsion,
- $\hat{V}_{eN}$ represents the electron–nucleus Coulomb attraction.

Explicitly, these terms can be written as:

$$\hat{T}_e = -\sum_i \frac{\hbar^2}{2m_e} \nabla^2_{r_i}, \tag{2}$$

$$\hat{T}_N = -\sum_I \frac{\hbar^2}{2M_I} \nabla^2_{R_I}, \tag{3}$$

$$\hat{V}_{ee} = \sum_{i<j} \frac{e^2}{4\pi|r_i - r_j|}, \tag{4}$$

$$\hat{V}_{NN} = \sum_{I<J} \frac{e^2 Z_I Z_J}{4\pi|R_I - R_J|}, \tag{5}$$

$$\hat{V}_{eN} = -\sum_{i,I} \frac{e^2 Z_I}{4\pi|r_i - R_I|}. \tag{6}$$

Here, r_i denotes the position of the i-th electron and R_I the position of the I-th nucleus, m_e is the electron mass, M_I the mass of nucleus I, Z_I its atomic number, e the elementary charge, and $\hbar$ the reduced Planck constant.

In practice, due to the large mass difference between nuclei and electrons, the Born–Oppenheimer approximation (the assumption that the wave functions of atomic nuclei and electrons in a molecule can be treated separately) is commonly employed. This approximation neglects the nuclear kinetic energy term and treats the nuclei as fixed classical point charges. The Hamiltonian then reduces to the purely electronic Hamiltonian:

$$\hat{H}_{el} = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{eN} + E_{NN}, \tag{7}$$

where E_{NN} is a constant corresponding to the nucleus–nucleus repulsion energy for a fixed molecular geometry.

However, this first-quantized form depends explicitly on the continuous electronic coordinates and does not naturally incorporate spin-orbital occupation. In order to make the problem compatible with quantum computational frameworks, the Hamiltonian is reformulated in second quantization. By expanding the electronic field operators in a chosen spin-orbital basis $\{\phi_i\}$, the Hamiltonian becomes:

$$\hat{H}_{el} = \sum_{ij} h_{ij} a_i^\dagger a_j + \frac{1}{2} \sum_{ijkl} h_{ijkl} a_i^\dagger a_j^\dagger a_k a_l, \tag{8}$$

where $a_i^\dagger$ and a_i are fermionic creation and annihilation operators. The coefficients h_{ij} correspond to one-electron integrals (kinetic energy and electron–nucleus interaction), while h_{ijkl} correspond to two-electron Coulomb integrals.

This second-quantized formulation naturally introduces occupation-number representation, which allows a direct mapping onto qubit states for quantum simulation.

2.1 STO 3G basis

The natural following question is: which type of basis functions should be chosen in order to accurately capture the behavior of electrons in atoms and molecules?

From quantum mechanics, we know that the exact hydrogenic $1s$ orbital is represented by a Slater-Type Orbital (STO) of the form:

$$\psi_{1s}(r) \propto e^{-\zeta r} \tag{9}$$

Where:

- ζ is the effective nuclear charge parameter.
- r is the radial distance from the nucleus to the electron, defined as $r = ||\mathbf{r}_i - \mathbf{R}_I||$.

However, despite their physical accuracy, Slater functions are not computationally convenient. The analytical evaluation of multi-electron integrals involving exponential terms of the form $e^{-\zeta r}$ is extremely costly, especially for two-electron Coulomb integrals.

For this reason, quantum chemistry commonly employs Gaussian-Type Orbitals (GTOs), which take the form

$$g(r) = e^{-\alpha r^2}, \tag{10}$$

where α is a positive parameter controlling the width of the Gaussian. Although Gaussians do not reproduce the exact cusp behavior at the nucleus and decay too rapidly at large distances, they possess a major computational advantage: products of Gaussian functions centered on different nuclei remain Gaussian. This property allows analytical evaluation of all required integrals.

In order to combine physical realism with computational efficiency, contracted Gaussian basis sets are introduced. In particular, the STO-3G basis set (Slater-Type Orbital approximated by 3 Gaussians) represents each atomic orbital as a linear combination of three primitive Gaussian functions:

$$\phi_{\text{STO-3G}}(r) = \sum_{k=1}^{3} c_k \, e^{-\alpha_k r^2}. \tag{11}$$

The coefficients c_k and exponents α_k are chosen such that this linear combination best approximates the corresponding Slater orbital.

Consequently, within a chosen basis set such as STO-3G, any molecular orbital $\psi(r)$ is expressed as a linear combination of these contracted Gaussian basis functions:

$$\psi(r) = \sum_i C_i \phi_i(r). \tag{12}$$

This expansion transforms the continuous Schrödinger equation into a finite-dimensional matrix eigenvalue problem, making numerical simulation possible.

3 Encoding and Jordan Wigner Mapping

As a result, once the electronic Hamiltonian has been projected onto a finite-dimensional space using a chosen basis set, the problem becomes algebraic rather than continuous. The infinite-dimensional Schrödinger equation is reduced to a finite-dimensional operator acting on a discrete Hilbert space spanned by spin-orbitals.

In order to perform quantum computation, we must now choose a representation of these spin-orbital states that is compatible with qubit systems. A natural choice is the occupation-number representation (also called Fock representation), where each spin-orbital is associated with a binary variable.

In this representation, if a spin-orbital i is occupied by an electron, we assign the value 1, and if it is unoccupied, we assign the value 0. Therefore, a many-electron state is represented as

$$|n_0, n_1, \ldots, n_{M-1}\rangle, \tag{13}$$

where $n_i \in \{0, 1\}$ and M is the total number of spin-orbitals.

For example, in the case of the H_2 molecule described in the STO-3G basis, there are two hydrogen atoms. Each hydrogen contributes one spatial orbital (the $1s$ orbital), and each spatial orbital gives rise to two spin-orbitals (spin-up and spin-down). Consequently, the system contains four spin-orbitals in total.

Thus, the electronic Hilbert space can be encoded using four qubits, where each qubit represents the occupation of one spin-orbital:

$$|n_0 n_1 n_2 n_3\rangle, \tag{14}$$

with $n_i = 1$ if the i-th spin-orbital is occupied and $n_i = 0$ otherwise.

3.1 Fermionic Operators and Anticommutation

In second quantization, electrons are described using fermionic creation and annihilation operators. Since electrons are fermions, these operators must satisfy the canonical anticommutation relations (CAR):

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0, \tag{15}$$

where δ_{ij} denotes the Kronecker delta.

The annihilation operator a_i removes an electron from spin-orbital i, while the creation operator $a_i^\dagger$ creates an electron in spin-orbital i. Acting on a Fock basis state

$$|n_0, n_1, \ldots, n_i, \ldots\rangle, \tag{16}$$

with $n_k \in \{0, 1\}$, these operators are defined as:

$$a_i|n_0, \ldots, n_i, \ldots\rangle = (-1)^{\sum_{k<i} n_k}\, n_i\, |n_0, \ldots, 0, \ldots\rangle, \tag{17}$$

$$a_i^\dagger|n_0, \ldots, n_i, \ldots\rangle = (-1)^{\sum_{k<i} n_k}\, (1 - n_i)\, |n_0, \ldots, 1, \ldots\rangle. \tag{18}$$

The phase factor $(-1)^{\sum_{k<i} n_k}$ ensures the antisymmetry of the fermionic wavefunction and guarantees that fermionic operators anticommute.

However, quantum computers manipulate qubits using Pauli operators, which commute when acting on different qubits. Therefore, a direct encoding of fermionic operators would not preserve the required anticommutation relations. To overcome this issue, the Jordan–Wigner transformation is introduced.

Jordan–Wigner Mapping

The Jordan–Wigner transformation maps fermionic operators onto qubit operators as follows:

$$a_i = \left(\prod_{k<i} Z_k\right) \frac{X_i + iY_i}{2}, \qquad a_i^\dagger = \left(\prod_{k<i} Z_k\right) \frac{X_i - iY_i}{2}. \tag{19}$$

The product of Pauli Z operators acts as a parity operator. Since

$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle, \tag{20}$$

the Z string computes the parity factor $(-1)^{\sum_{k<i} n_k}$, thus reproducing the correct fermionic sign structure. The operators

$$\sigma_i^+ = \frac{X_i - iY_i}{2}, \qquad \sigma_i^- = \frac{X_i + iY_i}{2}, \tag{21}$$

act as raising and lowering operators on qubit i:

$$\sigma_i^+|0\rangle = |1\rangle, \quad \sigma_i^+|1\rangle = 0, \tag{22}$$

$$\sigma_i^-|1\rangle = |0\rangle, \quad \sigma_i^-|0\rangle = 0. \tag{23}$$

Therefore, σ_i^+ creates an occupation on qubit i, while σ_i^- removes it. The Z string ensures the correct fermionic anticommutation relations, while the $\sigma^\pm$ operators implement the local occupation change.

Hamiltonian in Terms of Pauli Operators

Once the fermionic operators are mapped to qubit operators, the electronic Hamiltonian becomes a linear combination of Pauli strings:

$$H = \sum_{\alpha} c_\alpha P_\alpha, \tag{24}$$

where each P_α is a tensor product of Pauli matrices (I, X, Y, Z). This representation is directly compatible with quantum hardware, allowing the implementation of quantum algorithms such as the Variational Quantum Eigensolver (VQE).

4 Initialization and Hartree–Fock Method

4.1 Quantum State and Energy Level

Before describing the architecture of our solution, it is important to clarify the physical objective of our study. Our goal is to determine the ground state of the H_2 molecule. This requires solving the molecular Schrödinger equation, which is computationally demanding due to the interacting many-electron nature of the system.

More precisely, the Schrödinger equation defines an eigenvalue problem of the form

$$\hat{H}\Psi = E\Psi,$$

Among all possible solutions, we are particularly interested in the lowest eigenvalue E_0, corresponding to the ground state of the molecule. Physically, this state represents the most stable configuration of the system, as isolated quantum systems naturally relax toward their minimal energy state.

However, solving this eigenvalue problem exactly becomes intractable as the system size grows. For this reason, approximate and variational approaches are required. Before applying the Variational Quantum Eigensolver (VQE) algorithm, which we will introduce later, it is necessary to provide an initial approximation of the electronic wavefunction.

This initial guess is obtained through a classical Hartree–Fock calculation. The Hartree–Fock method approximates the many-electron wavefunction by a single Slater determinant constructed from optimized molecular orbitals. Although this approximation neglects electronic correlation effects, it provides a physically meaningful and computationally efficient reference state.

This Hartree–Fock state serves as the starting point (ansatz initialization) for the VQE algorithm. Providing a physically motivated initial state significantly improves convergence, as the optimization procedure begins in a region of Hilbert space already close to the true ground state.

4.2 Hartree–Fock Approximation

As stated previously, the main assumption underlying the Hartree–Fock method is that the many-electron wavefunction can be approximated by a single Slater determinant constructed from one-electron orbitals.

If we denote by $\phi_1, \phi_2, \ldots, \phi_N$ a set of orthonormal spin-orbitals (in the Hilbert space sense), a naive product of these orbitals would not satisfy the antisymmetry requirement imposed by the Pauli exclusion principle. Since electrons are fermions, the total wavefunction must change sign under the exchange of two particles.

To enforce this antisymmetry condition, the Hartree–Fock method constructs the wavefunction as a Slater determinant:

$$\Psi_{HF}(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(x_1) & \phi_2(x_1) & \ldots & \phi_N(x_1) \\ \phi_1(x_2) & \phi_2(x_2) & \ldots & \phi_N(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(x_N) & \phi_2(x_N) & \ldots & \phi_N(x_N) \end{vmatrix}.$$

Here, $x_i = (r_i, s_i)$ denotes both spatial and spin coordinates of electron i. The normalization factor $1/\sqrt{N!}$ ensures that the wavefunction is properly normalized.

4.2.1 Solving the Eigenvalue Problem with the Hartree–Fock Method

We assume that the spin-orbitals forming the Slater determinant are orthonormal. Under this assumption, the many-electron Schrödinger equation can be reformulated using the variational principle. Instead of directly solving the eigenvalue equation. We consider the expectation value of the Hamiltonian,

$$E = \langle \Psi | \hat{H} | \Psi \rangle,$$

which represents the total electronic energy of the system. According to the variational principle, this quantity is always greater than or equal to the exact ground-state energy. Therefore, minimizing this functional provides an approximation to the lowest eigenvalue of the Hamiltonian.

The problem is thus transformed into a constrained optimization problem: we seek the set of orthonormal spin-orbitals that minimizes the total energy, subject to the orthonormality conditions

$$\min_{\{\varphi_i\}_{i=1}^N} \quad E_{\mathrm{HF}}(\varphi_1, \ldots, \varphi_N)$$

$$\text{subject to} \quad \begin{cases} \langle \varphi_i, \varphi_j \rangle = \delta_{ij}, & \forall\, i, j \in \{1, \ldots, N\}, \\ \varphi_i \in \mathcal{H}, & \forall\, i \in \{1, \ldots, N\}, \end{cases}$$

To incorporate these constraints, we introduce a Lagrangian functional with matrix-valued Lagrange multipliers. Taking the functional derivative of this Lagrangian with respect to the spin-orbitals and imposing stationarity conditions leads to the Hartree–Fock equations.

These equations have the structure of an effective one-electron eigenvalue problem involving the Fock operator. Since the Fock operator itself depends on the occupied orbitals, the resulting equations are nonlinear and must be solved iteratively using the self-consistent field (SCF) procedure.

For simplicity, we omit the full derivation and focus on the conceptual framework underlying the method.

5 VQE Algorithm and Energy Minimization

5.1 Principle

After obtaining the initial reference state from the Hartree–Fock calculation, we apply the Variational Quantum Eigensolver (VQE) algorithm in order to approximate the ground-state energy of the molecular Hamiltonian.

As discussed previously, the electronic Hamiltonian has been mapped onto a qubit operator and expressed as a linear combination of Pauli strings:

$$\hat{H} = \sum_k c_k P_k,$$

where $c_k \in \mathbb{R}$ are coefficients and P_k are tensor products of Pauli operators.

We then introduce a parametrized quantum circuit (ansatz) depending on a set of real parameters θ. This circuit prepares a trial quantum state:

$$|\psi(\theta)\rangle = U(\theta)|\psi_{HF}\rangle,$$

where $|\psi_{HF}\rangle$ is the Hartree–Fock reference state.

Using the variational principle, we compute the expectation value of the Hamiltonian:

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle = \sum_k c_k \langle \psi(\theta) | P_k | \psi(\theta) \rangle.$$

Each term $\langle P_k \rangle$ is estimated by repeatedly preparing the quantum state and measuring the corresponding Pauli string. Due to the probabilistic nature of quantum measurements, multiple shots are required to obtain

a reliable estimate.

The objective of the VQE algorithm is to minimize $E(\theta)$ with respect to the parameters θ. A classical optimization routine (e.g., gradient descent, COBYLA, or SPSA) updates the parameters iteratively in order to reduce the energy.

The process is repeated until convergence, typically when the variation of the energy between successive iterations falls below a predefined threshold.

5.2 What is the ansatz and why is it needed?

In VQE, the ansatz is a parametrized quantum circuit used to generate candidate wavefunctions for the molecule. It can be seen as a model family

$$\{|\psi(\theta)\rangle\}_{\theta\in\mathbb{R}^d},$$

where each parameter vector θ defines one possible quantum state.

Its role is to restrict the search space to states that are physically relevant and efficiently preparable on a quantum device. Instead of exploring the full Hilbert space (which grows exponentially with the number of qubits), VQE explores only the manifold generated by the chosen ansatz.

Operationally, the circuit starts from a reference state (here, Hartree–Fock) and applies layers of gates controlled by θ. During optimization, the classical optimizer updates θ, the quantum computer prepares the new state, and the energy is remeasured. This loop continues until the minimum reachable energy is found within that ansatz family.

Therefore, the ansatz is central to VQE performance: if it is too simple, it cannot represent the true ground state (large bias); if it is too deep, optimization and hardware execution become harder (higher cost and noise sensitivity). In this project, we used a UCCSD-inspired ansatz because it provides a good balance between chemical expressiveness and practical implementation for small molecules.

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where $|\psi_{HF}\rangle$ is the Hartree–Fock reference state.

Using the variational principle, we compute the expectation value of the Hamiltonian:

$$E(\theta) = \langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle = \sum_k c_k \langle\psi(\theta)|P_k|\psi(\theta)\rangle.$$

Each term $\langle P_k\rangle$ is estimated by repeatedly preparing the quantum state and measuring the corresponding Pauli string as said before.

The objective of the VQE algorithm is to minimize $E(\theta)$ with respect to the parameters θ. A classical optimization routine (e.g., gradient descent, COBYLA, or SPSA) updates the parameters iteratively in order to reduce the energy.

The process is repeated until convergence, typically when the variation of the energy between successive iterations falls below a predefined threshold.

5.3 Common ansatz families and a short comparison

An ansatz is the parameterized quantum circuit used by VQE to create trial states $|\psi(\theta)\rangle$. The optimizer changes the parameters θ to minimize energy. A good ansatz is expressive enough to model the ground state, but simple enough to optimize.

UCCSD

What is this ansatz? UCCSD is a chemistry-inspired ansatz based on single and double electronic excitations.

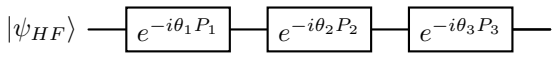

Figure 1: Simple UCCSD-style circuit: a product of parameterized excitation operators.

How to read the circuit: Each block $e^{-i\theta_k P_k}$ is a parameterized operation (from mapped excitation terms). The circuit starts from Hartree–Fock and adds controlled correlation step by step.

PUCCSD

What is this ansatz? PUCCSD is a paired version of UCCSD: it keeps the paired-excitation structure to reduce complexity.

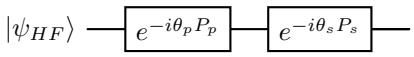

Figure 2: Simple PUCCSD-style circuit: paired doubles plus selected singles.

How to read the circuit: The first block encodes paired-double excitations, and the second block adds single excitations. This usually gives lower depth than full UCCSD while keeping good accuracy.

PUCCD

What is this ansatz? PUCCD is the most compact of the three here: it uses only paired-double excitations.

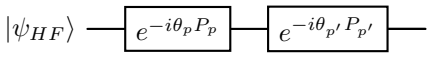

Figure 3: Simple PUCCD-style circuit: only paired-double excitation blocks.

How to read the circuit: Only paired-double terms are applied, so the circuit is shorter and often easier to optimize, especially for small systems like H_2.

6 Results analysis

To focus on algorithmic trade-offs, we fixed the molecular geometry at $d = 0.735$ Å (STO-3G basis) and evaluated all combinations of:

- 3 mappings: `Jordan--Wigner`, `Parity`, `Bravyi--Kitaev`,
- 2 optimizers: `SLSQP`, `COBYLA`,
- 3 ansatz: `UCCSD`, `PUCCSD`, `PUCCD`.

This gives 18 fixed-point VQE runs. For each run we measured total energy, absolute error, iterations, and runtime.

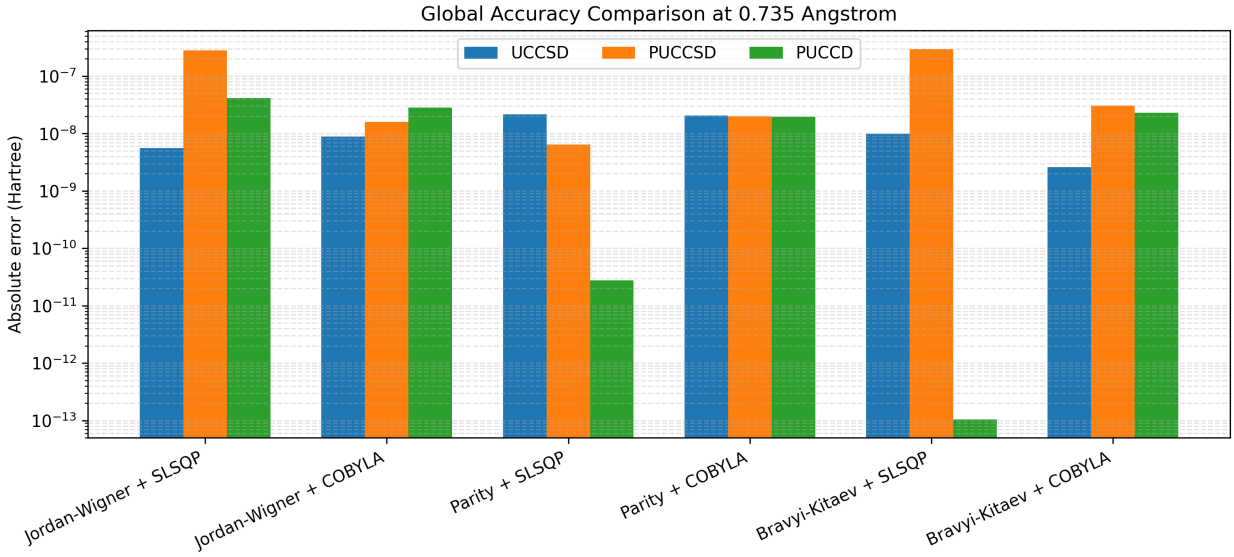

Figure 4: Global absolute-error comparison by mapping+optimizer combination, with the three ansatz shown together (log scale).

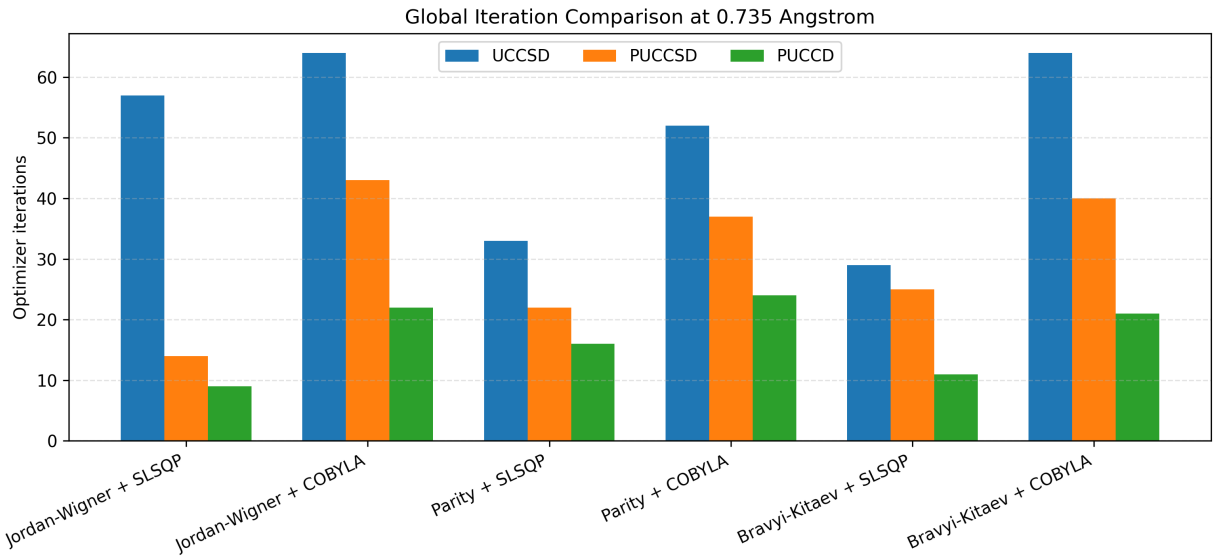

Figure 5: Global iteration comparison by mapping+optimizer combination, with the three ansatz shown together.

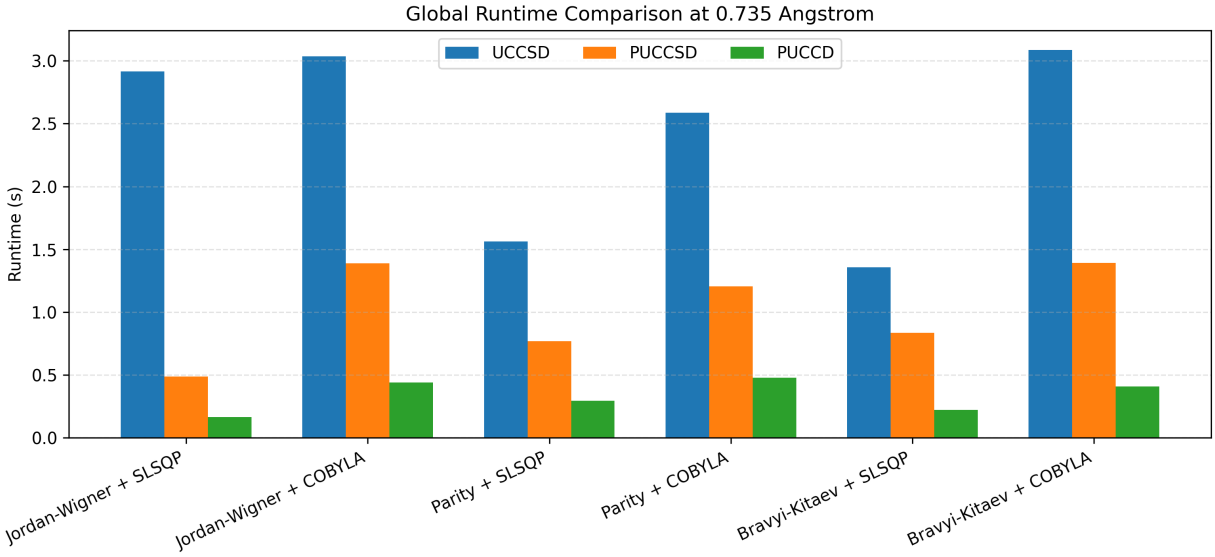

Figure 6: Global runtime comparison by mapping+optimizer combination, with the three ansatz shown together.

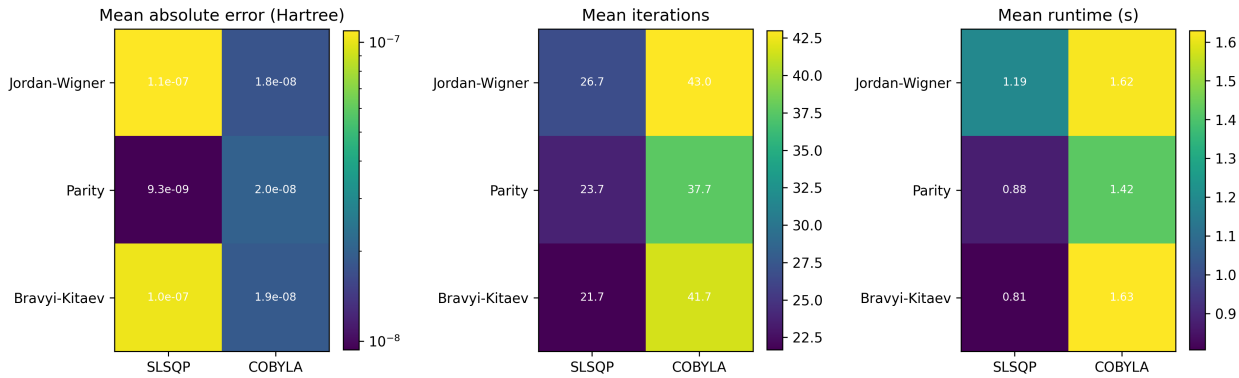

Figure 7: Mean metrics by mapping and optimizer (averaged over the three ansatz).

Main observations (mapping and optimizer focus)

All 18 runs remain close to the exact energy, but convergence behavior differs clearly by mapping/optimizer choice. The best single accuracy is obtained with `PUCCD + Bravyi--Kitaev + SLSQP` (1.04×10^{-13} Hartree), while the largest error is `PUCCSD + Bravyi--Kitaev + SLSQP` (2.97×10^{-7} Hartree), as seen in Figure 4.

From Figure 7, the optimizer effect is strong:

- `SLSQP` is faster and uses fewer iterations on average (about 24.0 iterations, 0.96 s),
- `COBYLA` is slower and needs more iterations (about 40.8 iterations, 1.56 s).

For accuracy, the trend is inverted in this benchmark: `COBYLA` has lower mean absolute error across all mappings/ansatz, while `SLSQP` shows larger error variability because of a few PUCCSD outliers.

The mapping effect is smaller than the optimizer effect for runtime/iterations, but still visible for accuracy:

- `Parity` gives the lowest mean absolute error overall (about 1.47×10^{-8} Hartree),
- `Jordan--Wigner` and `Bravyi--Kitaev` have similar average cost, with slightly higher mean errors (both around 6×10^{-8} Hartree in this dataset).

Role of the three ansatz in this mapping/optimizer comparison

The three ansatz remain important because they modulate each mapping/optimizer combination:

- `UCCSD`: most stable accuracy level, but highest optimization cost (mean ≈ 49.8 iterations, ≈ 2.42 s),
- `PUCCSD`: intermediate cost, but highest error dispersion due to several 10^{-7}-level cases,
- `PUCCD`: best efficiency (mean ≈ 17.2 iterations, ≈ 0.34 s) with very competitive accuracy.

Overall, the global plots make the trade-off clearer: mapping and optimizer choices set the convergence profile, while ansatz choice controls how costly and robust each profile becomes.

7 Conclusion

This project established a complete and reproducible pipeline for quantum simulation of molecular electronic structure, from the second-quantized Hamiltonian of H_2 to variational ground-state energy estimation with VQE. Using PySCF for integral generation and Qiskit for qubit encoding and optimization, we compared several fermion-to-qubit mappings and classical optimizers under the same physical setup.

At the equilibrium distance $d = 0.735$ Å, all tested workflows produced energies very close to the exact reference, with absolute errors spanning roughly 10^{-13} to 10^{-7} Hartree depending on the configuration. This confirms that, for small systems, VQE can reproduce chemically meaningful results with high precision. However, the experiments also showed that mapping and optimizer choices strongly affect convergence speed, iteration count, runtime, and error dispersion.

Overall, the study demonstrates that hybrid quantum-classical methods are already effective for benchmark molecules and useful for analyzing algorithmic trade-offs. Future work should target larger active spaces, improved ansatz design, measurement-cost reduction strategies, and noisy-hardware validation to evaluate how this approach scales beyond proof-of-concept systems.

Appendix: Supplementary distance validation

To ensure that the fixed-point study remains physically consistent with the molecular energy landscape, we include a small supplementary distance sweep using one reference workflow (`Jordan--Wigner + SLSQP + UCCSD`) over $d \in \{0.50,\ 0.735,\ 1.00,\ 1.50\}$ Å. As shown in Figure 8, the VQE points follow the exact curve trend, which supports the validity of the fixed-distance comparison used in the main analysis.

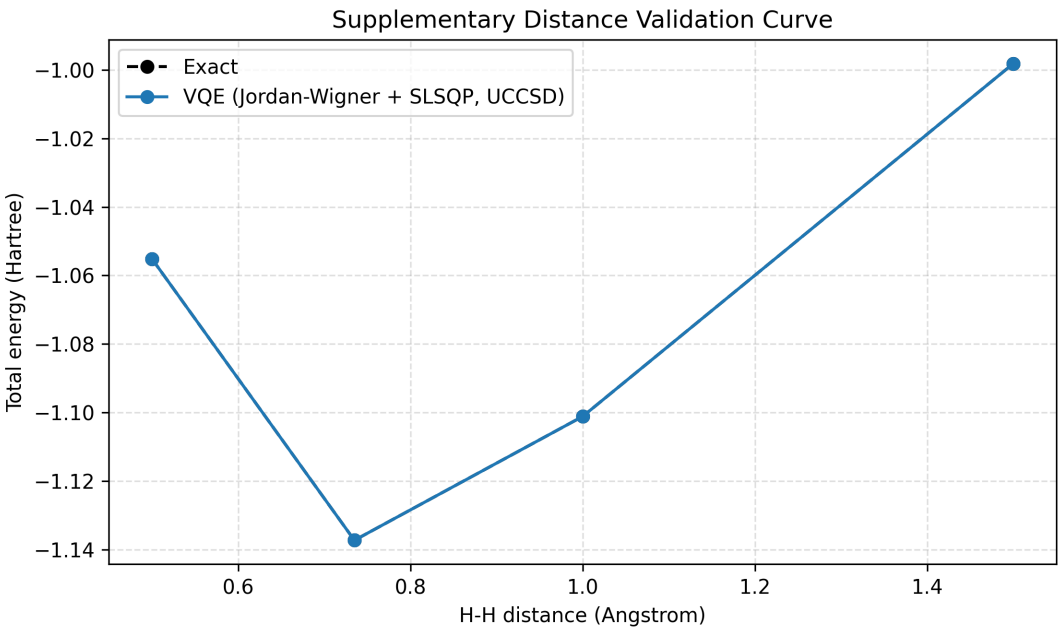

Figure 8: Supplementary validation curve: exact and VQE energies versus distance for a single reference workflow.

8 Annex

Algorithm 1: Variational Quantum Eigensolver (VQE) for Ground-State Energy Estimation

Data: Qubit Hamiltonian $\hat{H} = \sum_{k=1}^{M} c_k P_k$ (Pauli strings P_k and coefficients c_k);
Reference state $|\psi_{\mathrm{HF}}\rangle$ (Hartree–Fock initial state);
Ansatz circuit family $U(\theta)$ with parameters $\theta \in \mathbb{R}^d$;
Initial parameters $\theta^{(0)}$;
Classical optimizer `Opt` (e.g., COBYLA, SPSA, Gradient Descent);
Number of shots per measurement N_{shots};
Maximum iterations $I_{\max}$;
Convergence threshold ε (energy tolerance);
Optional: grouping strategy $\mathcal{G}$ for commuting Pauli strings.
Input: Backend configuration B (simulator or hardware; noise model if any).
Output: Approximate ground-state energy $E^\star$ and optimized parameters $\theta^\star$.

```
θ ← θ^(0)
E_prev ← +∞
// Optional: reduce measurement cost by grouping commuting Pauli strings
{S_1, ..., S_G} ← GroupPaulis({P_k}_{k=1}^M, G)      // S_g = set of simultaneously measurable terms
for i ← 1 to I_max do
    // 1) Prepare the variational quantum state
    |ψ(θ)⟩ ← PrepareState(U(θ), |ψ_HF⟩, B)
    // 2) Estimate the energy by measuring Pauli strings
    E(θ) ← 0
    for g ← 1 to G do
        // Measure all Pauli strings in group S_g with the same measurement basis
        Execute the circuit preparing |ψ(θ)⟩ and apply basis changes for S_g
        Collect N_shots measurement outcomes
        foreach P_k ∈ S_g do
            ⟨P̂_k⟩ ← EstimateExpectation(P_k, outcomes)          // sample mean of ±1 values
            E(θ) ← E(θ) + c_k · ⟨P̂_k⟩
    // 3) Check convergence
    if |E(θ) − E_prev| < ε then
        break
    E_prev ← E(θ)
    // 4) Classical optimization step (update parameters)
    θ ← OptimizerUpdate(Opt, θ, E(θ))          // or use gradient estimates if required
E* ← E(θ)
θ* ← θ
return (E*, θ*)
```

Quantum Simulation of the Electronic Structure of H_2